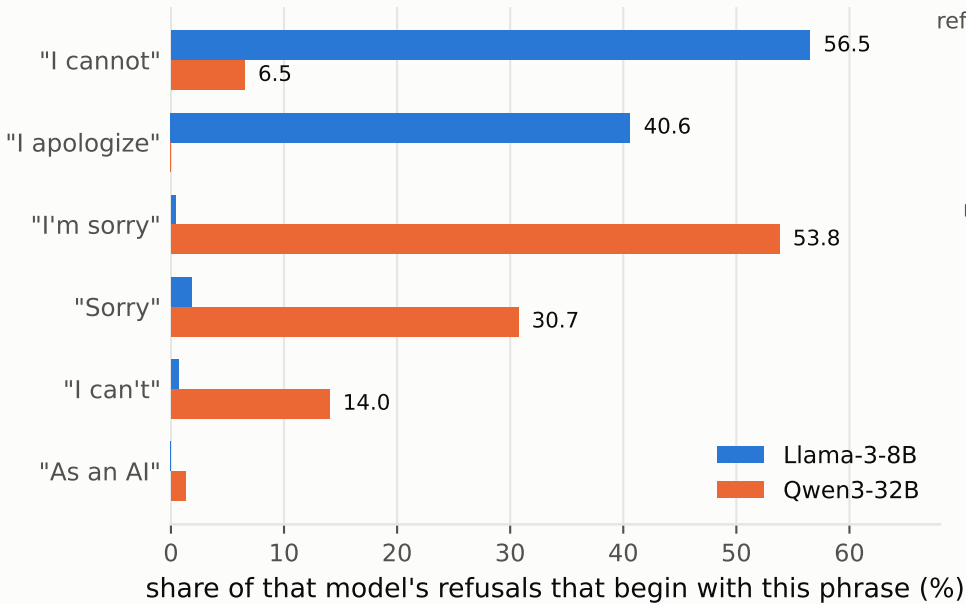


Two measurement choices the rest of the paper depends on

Left: openers mined from 23,595 Llama and 19,952 Qwen generations. "I cannot" opens 56.5% of Llama's refusals and 6.5% of Qwen's, so a phrase detector does not transfer between models. Right: on the full set all three signals look interchangeable; restricting to rewrites of the SAME original degrades the activation signals and leaves the logit intact. Qwen's probe coincides with its vector because the fit places all weight on one layer.

Refusal openers are model-specific



Controlling for topic separates the signals

